

Decision-Aware Physics-Informed Forecasting and Metaheuristic Allocation for NHS Critical-Care Surge Capacity Under Demand Uncertainty

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Abstract. Epidemic waves create rapid, spatially uneven pressure on critical-care services, requiring health systems to decide how to pre-position surge capacity and route patients under forecast uncertainty. We propose a forecast-to-decision pipeline that couples a per-region physics-informed neural forecaster on the refined $SEI_a I_s HCRD$ compartmental system to a robust mixed-integer-linear allocation model. The forecaster is trained directly on a multi-quantile pinball loss, which is decision-aware by construction (the $q^{(0.9)}$ component penalises under-prediction nine-to-one), and uses a learnable level-and-trend anchor that residual-connects the most recent observation to every horizon-quantile prediction. Quantile forecasts are discretised into three demand scenarios that drive a regional bed-surge allocation problem solved exactly via PuLP+CBC and approximately via a Genetic Algorithm, the Non-dominated Sorting Genetic Algorithm II, and Simulated Annealing on a shared LP-slave fitness oracle. On a seven-region NHS England case study spanning 1 August 2020 to 31 August 2022, the proposed forecaster reduces 14-day RMSE by 19% over per-origin re-fitted ARIMA and the 28-day under-estimation rate by four-fold; the tail-risk robust MILP achieves the lowest transfer burden in the policy set ($4.6\times$ below population-proportional) at no cost in expected or worst-case coverage. A four-axis sensitivity sweep (forecaster, surge budget, CVaR weight, mutual-aid travel cap) clarifies when each control binds.

Keywords: physics-informed neural networks, decision-aware forecasting, pinball loss, robust optimisation, metaheuristics, NSGA-II, critical-care planning, NHS England

1 Introduction

The COVID-19 pandemic exposed the vulnerability of national health systems to rapid, spatially uneven surges in critical-care demand. In England, sustained demand for adult critical-care beds and mechanical ventilation during successive waves required emergency capacity expansion and inter-region patient transfers. Decisions of this scale are made under deep uncertainty: planners must balance

the cost of expanding surge capacity against the risk of unmet demand, while keeping patient-transfer burden manageable across the seven NHS England regions. The computational intelligence community has produced strong tools for both halves of this problem — epidemic and hospital-demand forecasting [?, ?, ?] and robust facility-location and resource allocation under uncertainty [?, ?, ?] — but few works connect the two into a single decision-focused pipeline tailored to a national health system. The present work occupies the *temporal, per-region* corner of the forecasting design space and the *multi-method comparative* corner of the allocation design space; the spatiotemporal-graph alternatives surveyed in Section 2 are explicitly deferred to future work.

Existing forecasting models, including physics-informed neural networks (PINNs) [?, ?] and graph-based architectures such as STAN [?] and HOIST [?], are typically optimised for statistical accuracy alone. Yet for critical-care planning the operational cost of forecast errors is asymmetric: underestimating demand by ten beds risks patients without ventilators, while overestimating wastes prepared capacity. Standard squared-error or absolute-error objectives treat these symmetrically. Conversely, most optimisation formulations for pandemic capacity planning assume demand is deterministic or drawn from off-the-shelf statistical forecasts [?, ?], leaving unexploited the dynamics-aware information that physics-informed forecasters can provide. Recent work begins to close this gap: Liu and Cao [?] couple spatiotemporal graph attention with multi-objective optimisation for vaccine and non-pharmaceutical-intervention allocation, and Shams Eddin and El Hajj [?] integrate SEIRD-driven demand into robust hospital-bed scheduling for a single US region. To our knowledge no published work combines (a) physics-informed regional epidemic forecasting, (b) decision-aware training that explicitly penalises operationally costly forecast errors, and (c) metaheuristic robust optimisation for NHS England critical-care surge capacity.

We close that gap with a forecast-to-decision pipeline organised in four phases. *Phase A* is a per-region PINN-SEIRD module that produces multi-horizon quantile forecasts of mechanical-ventilation bed occupancy. The PINN is first pre-trained to satisfy the refined $SEI_a I_s HCRD$ ODE system, producing twelve time-varying physics-informed features (eight compartmental states and four transmission/clinical parameters) that are then fed — frozen — into a temporal GRU head. The head emits the 10th, 50th and 90th predictive quantiles per horizon *directly* via a multi-quantile *pinball loss* (decision-aware by construction, since the $q^{(0.9)}$ component asymmetrically penalises under-prediction nine-to-one), with no post-hoc Monte-Carlo sampling. A learnable level-and-trend anchor skip-connects the most recent observation and recent weekly slope to every horizon-quantile prediction, trading short-horizon flexibility for long-horizon stability. *Phase B* discretises the quantile forecasts into three demand scenarios (low/median/high) with probabilities $\pi = (0.20, 0.60, 0.20)$. *Phase C* solves a regional bed-surge allocation problem using those scenarios, comparing a deterministic mixed-integer linear programme, a tail-risk robust MILP, a Genetic Algorithm with LP-slave decomposition, the Non-dominated Sorting Genetic Algorithm II [?], Simulated Annealing, and four closed-form heuristic baselines.

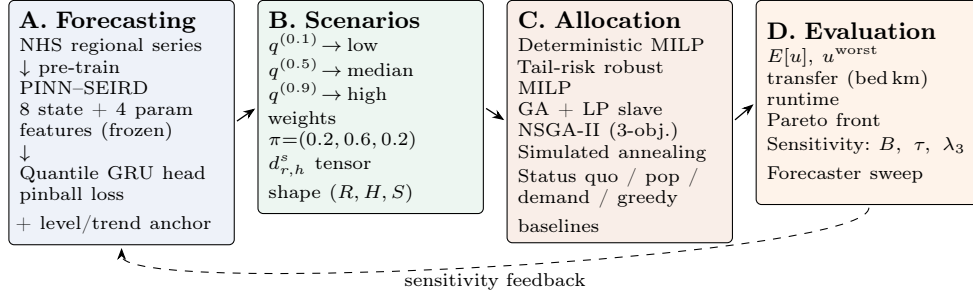


Fig. 1. Four-phase forecast-to-decision pipeline. Phase A pre-trains a per-region PINN-SEIRD on the refined $SEI_a I_s HCRD$ system and feeds its frozen state-and-parameter features into a quantile GRU head trained with a multi-quantile pinball loss. Phase B discretises the $q^{(0.1)}/q^{(0.5)}/q^{(0.9)}$ forecasts into a three-scenario demand tensor. Phase C compares an exact MILP, a tail-risk robust MILP, three metaheuristics, and four heuristic baselines on the same problem instance. Phase D reports operational and Pareto-front metrics and sweeps the surge budget B , the CVaR weight λ_3 , the travel-time cap τ , and the upstream forecaster.

Phase D reports operational metrics (expected unmet, worst-case unmet, transfer burden, runtime) plus sensitivity to the surge budget, the CVaR weight, and the mutual-aid travel-time cap.

Our contributions are: (1) a forecast-to-decision pipeline reproducible on open NHS data, with all four phases released as installable Python console commands; (2) a *decision-aware* multi-quantile pinball training objective coupled with a level-and-trend anchor skip connection, jointly responsible for a 19% RMSE reduction over per-origin re-fitted ARIMA at the operational 14-day horizon and a four-fold reduction in underestimation rate at $h = 28$; (3) a hybrid-regional robust allocation model with explicit MILP/GA/NSGA-II/SA comparison, revealing that the robust MILP achieves the lowest transfer burden in the policy set ($4.6\times$ below population-proportional, at identical expected and worst-case unmet) and that the metaheuristics match it within a few percent at 20–115 s runtime; (4) a refined $SEI_a I_s HCRD$ compartmental model that separates the hospitalisation ratio ω ($I_s \rightarrow H$) from the critical-care escalation proportion ϕ ($H \rightarrow C$), correcting an over-loading of ω in earlier work [?]; and (5) an NHS England case study spanning five forecasters — PinnGRU, ARIMA, per-region GRU, XGBoost, and seasonal-naive(7) — plus a three-axis sensitivity sweep over surge budget, CVaR weight, and travel-time cap, quantifying when forecast quality versus parameter choice dominates the operational outcome. Figure 1 summarises the four-phase pipeline. Section 2 reviews related work; Sections 3 and 4 describe the forecasting and optimisation modules; Section 5 presents the case study; Sections 6 to 8 report results, discuss limitations, and conclude.

2 Related Work

The closest precedent for our pipeline is Bertsimas et al. [?], who couple an extended DELPHI compartmental model of COVID-19 to a bilinear mixed-integer program for the location of one hundred mass-vaccination sites across US states, solving the resulting non-convex problem by a coordinate-descent scheme that alternates forward simulation of the discretised dynamics with a linear-programming approximation of the bilinear constraints. Their pipeline is the methodological parallel to ours; the differences are domain (vaccination-site siting versus critical-care surge), coupling (iterative joint optimisation versus our decoupled forecast-then-optimise), and forecasting objective (mean-squared error on DELPHI states versus our decision-aware multi-quantile pinball loss). Our framework also draws on the formal Smart Predict-then-Optimize framework of Elmachtoub and Grigas [?] and on the neural-ODE-based equitable opioid facility-location work of Luo and Stellato [?], who fit per-state SIR-like models via a neural-ODE-inspired procedure and embed the discretised dynamics into a McCormick-relaxed MIP. We adopt multi-quantile training in the spirit of the COVID-19 Forecast Hub’s Weighted Interval Score evaluation [?], which treats the upper quantile ($q^{0.9}$) as the operationally relevant safety bound. We review the three relevant subliterations below.

2.1 Epidemic and Hospital-Demand Forecasting

Statistical and recurrent baselines—ARIMA, Prophet, and LSTM/GRU architectures—remain competitive when applied per region [?]. Closest in target to our work, Valente et al. [?] forecast COVID-19 intensive-care occupancy in the Lazio region of Italy using a bidirectional LSTM with Google Community Mobility features lagged by twenty-one days following Cartení et al. [?], and exploit stochastic-ensemble Monte-Carlo dropout for prediction intervals; they report mean absolute error of approximately seven beds on weekly horizons. Graph neural networks add explicit spatial coupling: STAN [?] combines graph-attention pooling with an SIR-consistency loss for COVID-19 case prediction at US county and state level, and HOIST [?] introduces an Ising-dynamics-inspired regulariser over an adaptively-learned location graph for hospitalisation prediction across 2,299 US counties. Physics-informed neural networks embed compartmental ordinary differential equations directly into the training loss [?,?], supporting dynamics-consistent forecasts under regime shifts. NHS-specific work is sparse: Hunter et al. [?] target elective-surgery ICU demand at a single trust, and prior work by the present authors [?,?] forecasts national-scale mechanical-ventilation demand. None of these targets the multi-region NHS England critical-care planning problem, and all optimise statistical rather than operationally-aligned error metrics.

2.2 Robust Optimisation for Healthcare Resource Allocation

Pandemic capacity planning has produced a rich optimisation literature. Eriskin et al. [?] develop a robust multi-objective MILP for emergency-facility location

and patient assignment under demand uncertainty. Wan et al. [?] propose a two-stage multi-objective ICU bed allocation model solved by an improved NSGA-II on Lyon hospital data. Shams Eddin and El Hajj [?] integrate SEIRD-driven demand with standard beds, buffer beds, field hospitals, and inter-hospital transfers, validated on Northern Virginia data. In the vaccine supply-chain space, Kargar et al. [?] couple a Covasim agent-based simulator [?] to a multi-echelon bi-objective MILP solved by the augmented ε -constraint method, exposing the cost-versus-emissions Pareto front; we adopt the augmented ε -constraint paradigm in our allocation comparison, complementing exact MILP and NSGA-II with an exact non-dominated-solution generator. Tang et al. [?] apply adaptive large neighbourhood search to mobile-vaccination scheduling, exemplifying the broader operational-research-meets-public-health literature beyond static facility location. Metaheuristic methods—GA, NSGA-II [?], simulated annealing, and ALNS—are routinely employed when problem scale defeats exact MILP. The bulk of this literature is single-region or US-focused; trust-level allocation across NHS England regions and Integrated Care Boards has, to our knowledge, not been studied with the methodological breadth proposed here.

2.3 Forecast-to-Decision Pipelines

The integration of predictive forecasts into prescriptive optimisation is a maturing research area. Elmachtoub and Grigas [?] formalise the Smart Predict-then-Optimize framework with a loss aware of the downstream decision problem. In healthcare, Bertsimas et al. [?] provide the closest end-to-end precedent via coordinate-descent coupling of DELPHI simulation with bilinear MIP allocation; Luo and Stellato [?] provide an analogous predict-then-optimize architecture for opioid treatment-facility location with neural-ODE parameter estimation and McCormick-envelope relaxation; and Liu and Cao [?] couple spatiotemporal graph-attention forecasting to multi-objective vaccine and non-pharmaceutical-intervention allocation. Our framework adopts the predict-then-optimize structure, contributes a domain-specific decision-aware training objective that asymmetrically penalises operationally costly under-prediction, and broadens the optimisation comparison to include exact MILP, augmented ε -constraint, NSGA-II, GA, and simulated annealing. By treating the forecasting module as a substitutable component, the framework is forecasting-backend agnostic: alternative graph-based or transformer-based forecasters can be swapped in without changes to the allocation layer.

3 Forecasting Module

The forecasting module produces multi-horizon point and quantile estimates of critical-care demand for each NHS region. It comprises three parts: a per-region physics-informed neural epidemic estimator, a temporal forecasting head with a level-and-trend anchor that emits quantiles directly, and a pinball-loss training objective that operationalises decision-aware prediction without separate asymmetric penalty terms.

3.1 Refined SEIRD Compartmental Model

For each region $r \in \mathcal{R}$ with population N_r , we model the local epidemic using a refined eight-compartment $SEI_a I_s HCRD$ system. Let $U^r(t) = (S_r, E_r, I_{a,r}, I_{s,r}, H_r, C_r, R_r, D_r)$ denote susceptible, exposed, asymptomatic infectious, symptomatic infectious, hospitalised, critical-care, recovered, and deceased compartments respectively. Figure 2 shows the flow topology, in which the distinct probabilities ω ($I_s \rightarrow H$) and ϕ ($H \rightarrow C$) disambiguate the two clinical thresholds. The system evolves according to

$$\begin{aligned}
\dot{S}_r &= -\beta_r S_r (I_{s,r} + I_{a,r}) / N_r + \eta_r R_r \\
\dot{E}_r &= \beta_r S_r (I_{s,r} + I_{a,r}) / N_r - \alpha E_r \\
\dot{I}_{s,r} &= \alpha \rho E_r - d_s I_{s,r} \\
\dot{I}_{a,r} &= \alpha (1 - \rho) E_r - d_a I_{a,r} \\
\dot{H}_r &= d_s \omega I_{s,r} - d_H H_r - \mu H_r \\
\dot{C}_r &= \phi d_H H_r - \gamma_{c,r} C_r - \delta_{c,r} C_r \\
\dot{R}_r &= d_s (1 - \omega) I_{s,r} + d_a I_{a,r} + (1 - \phi) d_H H_r + \gamma_{c,r} C_r - \eta_r R_r \\
\dot{D}_r &= \mu H_r + \delta_{c,r} C_r,
\end{aligned} \tag{1}$$

subject to conservation $\sum_{\bullet} \bullet_r(t) = N_r$. Here α, d_s, d_a, d_H are *rates* with units of inverse-days (their reciprocals are the corresponding clinical periods), ρ is the symptomatic proportion, ω is the hospitalisation ratio of symptomatic cases, μ is the in-hospital mortality rate, and $\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r$ are time-varying region-specific parameters learned from data.

Refinement of the prior compartmental model. The earlier formulation in [?] overloaded the parameter ω to represent both the $I_s \rightarrow H$ hospitalisation ratio and, via $(1 - \omega)$, the $H \rightarrow C$ critical-care escalation. These quantities are biologically distinct: the first is determined by clinical severity at admission, the second by in-hospital deterioration. We separate them by introducing ϕ for the $H \rightarrow C$ escalation, leaving ω for hospitalisation alone. Equation (1) reflects this refinement.

3.2 Per-Region PINN-SEIRD

For each region r , the PINN consists of two fully-connected feedforward networks following [?]: a state network $U_{NN}^r : t \mapsto U^r(t)$ with five hidden layers of 20 units, and a parameter network $X_{NN}^r : t \mapsto (\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r)$ with three hidden layers of 20 units. Both use tanh hidden activations, sigmoid outputs (so all eight compartments and four parameters lie in $[0, 1]$), and Xavier initialisation. The PINN loss combines a data-fit term on observable compartments $\{H, C\}$ — the only ones for which NHS England publishes daily counts at regional resolution —

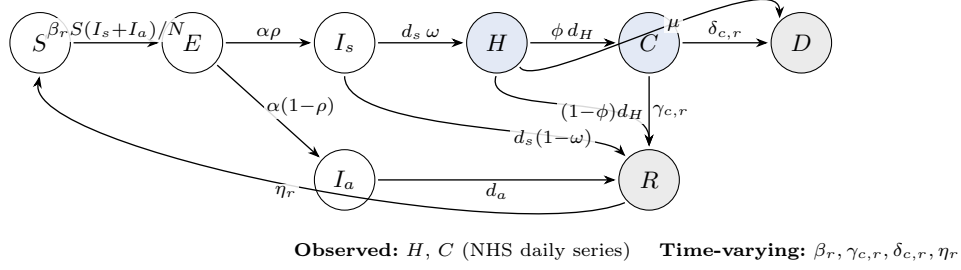


Fig. 2. Refined $SEI_a I_s H C R D$ compartmental model. The flow $E \rightarrow I_s \rightarrow H \rightarrow C$ carries the hospitalisation ratio ω and the critical-care escalation ϕ as *separate* parameters — the former determined by severity at admission, the latter by in-hospital deterioration. Shaded nodes (H, C) are observed compartments constraining the PINN loss; the terminal R and D nodes absorb non-observed flows. Region-specific time-varying parameters ($\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r$) are emitted by the PINN parameter network; the remaining rates are fixed clinical constants.

with a residual loss against Equation (1):

$$\mathcal{L}_r^{PINN} = \sum_{t \in \mathcal{T}_{\text{train}}} \sum_{k \in \{H, C\}} (U_{NN,k}^r(t) - y_{r,t,k}/N_r)^2 + \lambda_{ode} \sum_{t \in \mathcal{T}_{\text{coll}}} \left\| \frac{dU_{NN}^r(t)}{dt} - \mathcal{N}(U_{NN}^r(t); X_{NN}^r(t)) \right\|^2, \quad (2)$$

where $\mathcal{N}$ denotes the right-hand side of Equation (1), target observations are normalised by population, and derivatives are evaluated via PyTorch automatic differentiation at 256 collocation points sampled uniformly in $\mathcal{T}_{\text{coll}}$ at every iteration. We pre-train each region’s PINN for 1500 Adam steps with $\lambda_{ode} = 0.1$ and learning rate 5×10^{-3} ; final losses converge to $\mathcal{O}(10^{-6})$ on both terms, indicating tight fit to the observations *and* consistency with the ODE dynamics. The learned time-varying parameters are region-specific and non-constant (supplementary material); their predictive value is quantified by the *w/o PINN parameter features* ablation in Table 1.

3.3 Temporal Forecasting Head with Level-and-Trend Anchor

Once pre-trained, the PINN is *frozen*: its outputs are reused as a physics-informed temporal embedding for the forecaster. For each lookback window of length $L = 28$ days ending at the forecast origin t , the augmented feature sequence is

$$\mathbf{f}_{r,\tau} = [U_{NN}^r(\tau) \parallel X_{NN}^r(\tau) \parallel \tilde{y}_{r,\tau}] \in \mathbb{R}^{13}, \quad \tau \in \{t - L + 1, \dots, t\}, \quad (3)$$

where $\tilde{y}_{r,\tau} = (y_{r,\tau} - \bar{y}_r)/s_{y,r}$ is the z-scored observed target on the train+val period of region r . The sequence is processed by a two-layer GRU with hidden dimension 128 and dropout 0.15; the last hidden state $\mathbf{h}_{r,t}$ is decoded by a per-horizon two-layer MLP (width 64, GELU) into a triple of standardised quantile predictions $\Delta_{r,t}^{h,q} \in \mathbb{R}$ for $h \in \mathcal{H} = \{7, 14, 21, 28\}$ and $q \in \mathcal{Q} = \{0.1, 0.5, 0.9\}$.

To make the GRU’s job a residual one — predicting the *deviation* from a sensible baseline rather than the absolute level — we add a *level-and-trend anchor* skip connection:

$$\hat{z}_{r,t}^{h,q} = \Delta_{r,t}^{h,q} + \alpha^{h,q} \tilde{y}_{r,t} + \gamma^{h,q} s_{r,t}^{(7)} \left(\frac{h}{7}\right)^{\varphi^q}, \quad (4)$$

where $\tilde{y}_{r,t}$ is the most recent z-scored observation, $s_{r,t}^{(7)} = \tilde{y}_{r,t} - \tilde{y}_{r,t-7}$ is the recent one-week slope, and $\alpha^{h,q}, \gamma^{h,q}, \varphi^q \in (0, 1)$ are learnable scalars (parameterised by sigmoids on free logits). The exponent φ^q *dampens* the linear trend extrapolation per quantile, a Holt-Winters-style mechanism that prevents the slope-based projection from over-shooting at long horizons. We initialise $\varphi^{q=0.5} = \sigma(2) \approx 0.88$ (near-linear, accurate point forecasting) and $\varphi^{q=0.1} = \varphi^{q=0.9} = \sigma(0) = 0.5$ (square-root damping, tighter probabilistic bounds). All anchor weights are learnt jointly with the rest of the network during forecaster training. Point and interval forecasts in the original scale follow by inverting standardisation: $\hat{y}_{r,t+h-1}^q = \hat{z}_{r,t}^{h,q} s_{y,r} + \bar{y}_r$.

3.4 Decision-Aware Training via Pinball Loss

The forecaster is trained by minimising the multi-quantile pinball loss of Koenker and Bassett [?], in the form adopted by the COVID-19 Forecast Hub Weighted Interval Score framework [?]:

$$\mathcal{L}_{\text{pinball}} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbb{E}_{(t,h)} \left[\max(q(\tilde{y}_{r,t+h-1} - \hat{z}_{r,t}^{h,q}), (q-1)(\tilde{y}_{r,t+h-1} - \hat{z}_{r,t}^{h,q})) \right], \quad (5)$$

evaluated at all $(h, q) \in \mathcal{H} \times \mathcal{Q}$ pairs. Two operational properties are notable. First, the $q = 0.5$ pinball collapses to the absolute-error (L1) loss for the median, so it directly targets point accuracy without an added Huber term. Second, the $q = 0.9$ pinball is *asymmetric*: a forecast that falls below the realised value (an under-prediction, $y - \hat{y} > 0$) incurs loss $0.9(y - \hat{y})$, while a forecast that exceeds it incurs only $0.1(\hat{y} - y)$ — a nine-to-one penalty ratio favouring conservative upper-quantile estimates. This is the *decision-aware* component, taking the role that an explicit asymmetric hinge played in earlier compositions of the loss [?] but with the asymmetry built into a single strictly proper scoring rule rather than an ad-hoc additive term [?]. Reporting $\hat{y}^{0.5}$ as the point forecast and $[\hat{y}^{0.1}, \hat{y}^{0.9}]$ as the 80% predictive interval, the same model serves point and probabilistic users with no post-hoc Monte-Carlo sampling.

We train each region’s forecaster for up to 800 AdamW epochs ($\text{lr} = 10^{-3}$, weight decay 10^{-4} , cosine annealing), with the training set restricted to lookback windows whose forecast targets fall within the Alpha-wave training split. Early stopping monitors validation pinball loss over windows whose targets fall in the Delta-wave validation period (patience 80). Once selected, the model is frozen and applied to the Omicron test period via rolling-origin evaluation at $\Delta t = 7$ days, yielding 32 origin dates over the 9-month evaluation horizon.

4 Regional Allocation Model

The allocation module decides how many additional mechanical-ventilation beds to provide in each NHS region, and how excess demand should be redistributed to neighbouring regions through inter-regional patient transfers, given the quantile forecasts $\hat{q}^{(0.1)}, \hat{q}^{(0.5)}, \hat{q}^{(0.9)}$ produced in Section 3. We present a deterministic mixed-integer-linear programme, a tail-risk robust objective, and three meta-heuristic comparators.

4.1 Sets, Parameters and Decision Variables

The scope is *hybrid regional*: candidate sites coincide with the seven NHS England commissioning regions, so the demand-node and capacity-site indices collapse to a single set $\mathcal{R} = \{r_1, \dots, r_7\}$ with $|\mathcal{R}| = 7$. Every region is always “open” (no facility-opening binary), and per-bed surge cost is uniform across regions, so the model decides only how many surge beds to place in each region and how to route patients between them.

Indices are: regions $r, r' \in \mathcal{R}$; forecast horizons $h \in \mathcal{H} = \{7, 14, 21, 28\}$ days; demand scenarios $s \in \mathcal{S} = \{\text{low}, \text{median}, \text{high}\}$ corresponding to the $q^{(0.1)}, q^{(0.5)}, q^{(0.9)}$ quantile forecasts with weights $\pi_s = (0.20, 0.60, 0.20)$. Parameters are: forecast demand $d_{r,h}^s = \hat{q}_r^{(s)}(t_0 + h)$ at decision origin t_0 ; baseline capacity C_r (regional Delta-peak occupancy $\times 1.05$ safety margin); per-region surge cap $K_r = 0.5 C_r$; total surge budget $B = 0.20 \sum_r C_r$; great-circle inter-centroid distance $D_{rr'}$ in km (scaled by a 1.3 detour factor on the road network); inter-regional travel time $T_{rr'} = 60 D_{rr'} / 80$ km/h in minutes; and the mutual-aid travel-time cap $\tau = 240$ min (four hours). Decision variables are: surge-bed allocation $b_{r,h} \in [0, K_r]$; inter-regional transfer flow $z_{rr',h}^s \geq 0$, defined only for (r, r') pairs with $T_{rr'} \leq \tau$; and unmet demand $u_{r,h}^s \geq 0$. Auxiliary $W \geq 0$ tracks the upper-quantile tail in the robust extension.

4.2 Forecast-Driven Regional MILP

The deterministic model places surge capacity and routes patients to minimise unmet demand and transfer burden under the median forecast scenario. Let m denote the median scenario, corresponding to $q^{(0.5)}$:

$$\min \underbrace{\sum_{r,h} u_{r,h}^m}_{\text{unmet demand}} + \rho \underbrace{\sum_{r \neq r', h} D_{rr'} z_{rr',h}^m}_{\text{transfer burden}}, \quad (6)$$

where $\rho = 10^{-3}$ is a small weight that breaks ties towards allocations using fewer long-distance transfers without distorting the primary unmet-demand objective. The model is subject to demand satisfaction, capacity, expansion-cap, budget,

and travel-time constraints:

$$\sum_{r': T_{rr'} \leq \tau} z_{rr',h}^m + u_{r,h}^m \geq d_{r,h}^m \quad \forall r \in \mathcal{R}, h \in \mathcal{H}, \quad (7)$$

$$\sum_{r': T_{r'r} \leq \tau} z_{r'r,h}^m \leq C_r + b_{r,h} \quad \forall r \in \mathcal{R}, h \in \mathcal{H}, \quad (8)$$

$$b_{r,h} \leq K_r \quad \forall r \in \mathcal{R}, h \in \mathcal{H}, \quad (9)$$

$$\sum_r \max_h b_{r,h} \leq B, \quad (10)$$

$$z_{rr',h}^m = 0 \quad \text{if } T_{rr'} > \tau, \quad (11)$$

$$b_{r,h}, z_{rr',h}^m, u_{r,h}^m \geq 0. \quad (12)$$

Equation (7) ensures that the demand $d_{r,h}^m$ in region r at horizon h under the median scenario is either treated locally, received from elsewhere, or recorded as unmet. Equation (8) caps each region's inbound flow at its augmented capacity (baseline plus surge). Equation (9) enforces the per-region expansion cap. Equation (10) bounds the total surge investment in the peak-horizon demand profile (we use the peak because the same physical beds must serve every horizon up to and including peak load). Equation (11) suppresses transfers exceeding the four-hour mutual-aid window.

4.3 Tail-Risk Robust Objective

To explicitly hedge against the upper-quantile demand realisation ($s = \text{high}, q^{(0.9)}$), the robust model generalises the median formulation by imposing the demand and capacity constraints for every $s \in \mathcal{S}$ and adding a tail-risk penalty. Define the tail scenario subset $\mathcal{S}_{\text{tail}} = \{\text{high}\}$ and the auxiliary $W \geq \sum_{r,h} u_{r,h}^s$ for every $s \in \mathcal{S}_{\text{tail}}$. The robust objective is

$$\min \sum_{s \in \mathcal{S}} \pi_s \left(\sum_{r,h} u_{r,h}^s + \rho \sum_{r \neq r', h} D_{rr'} z_{rr',h}^s \right) + \lambda_3 W, \quad (13)$$

where $\lambda_3 = 0$ recovers the risk-neutral three-scenario objective and $\lambda_3 \rightarrow \infty$ yields a worst-case (min-max) policy on the high-demand scenario. We report results for $\lambda_3 = 1$, giving equal weight to expected shortage and high-scenario tail shortage.

4.4 Exact and Metaheuristic Solvers

For the regional formulation the model has 28 surge-bed variables, up to $7 \times 7 \times 4 \times 3 = 588$ transfer-flow variables, and $7 \times 4 \times 3 = 84$ unmet-demand variables, plus the tail auxiliary in the robust extension. We compare an exact MILP, three metaheuristics, and four naive baselines on the same problem instance.

Exact MILP via PuLP+CBC. The deterministic and robust models are formulated in PuLP and solved with the bundled CBC branch-and-cut solver. The deterministic model optimises only the median scenario by construction, and the achieved allocation is then re-evaluated by an LP slave over the full three-

scenario set to obtain honest expected- and worst-case unmet metrics. The robust model optimises all three scenarios with the $\lambda_3 = 1$ tail-risk term active. Both runs finish in well under a second on the seven-region instance.

Master-slave structure for the metaheuristics. For fixed peak surge allocation $b_r^* = \max_h b_{r,h}$, the residual problem in (b, z, u) becomes a linear programme on the transfer-flow and unmet-demand variables alone. Every metaheuristic operates on the continuous vector $\mathbf{b}^* \in \mathbb{R}_+^7$ subject to $\sum_r b_r^* \leq B$ and $b_r^* \leq K_r$, calling the LP slave once per fitness evaluation to compute the resulting expected unmet, transfer burden, and worst-case unmet.

Genetic Algorithm (single-objective). A pymoo Genetic Algorithm minimises expected unmet (plus the tie-break transfer-burden term) over $\mathbf{b}^*$ with real-valued encoding, SBX crossover ($\eta_c = 15$, $p_c = 0.9$), polynomial mutation ($\eta_m = 20$, $p_m = 1/7$), tournament selection, population size 50, 40 generations. A repair operator projects each candidate onto the budget simplex $\sum_r b_r^* \leq B$ after every genetic operation, following Resende and Werneck [?].

NSGA-II (three-objective). The Non-dominated Sorting Genetic Algorithm II [?] produces a Pareto front over three objectives evaluated simultaneously:

$$f_1(\mathbf{b}^*) = \sum_r b_r^*, \quad (\text{total surge beds, "cost"}) \quad (14)$$

$$f_2(\mathbf{b}^*) = \sum_{r,h,s} \pi_s u_{r,h}^{s*}, \quad (\text{expected unmet demand}) \quad (15)$$

$$f_3(\mathbf{b}^*) = \sum_{r \neq r', h, s} \pi_s D_{rr'} z_{rr',h}^{s*}, \quad (\text{transfer burden}) \quad (16)$$

where the starred quantities are LP-slave optima for the given $\mathbf{b}^*$. We use pymoo’s default NSGA-II operators with the same encoding, population, and generation count as the single- objective GA. A *representative point* is selected from the non-dominated set by the L2-distance-to-ideal-point rule, normalising each objective to $[0, 1]$ on the front.

Simulated Annealing comparator. A hand-rolled Simulated Annealing run uses geometric cooling ($T_0 = 1.0$, $\gamma = 0.97$) over Gaussian-perturbation neighbourhoods ($\sigma = 0.10 B$) with budget-simplex repair after every proposal, sharing the same LP-slave fitness oracle. The run length is 400 proposals.

Heuristic baselines. For completeness we include four closed-form allocation policies: *status quo* ($\mathbf{b}^* = 0$, no surge); *population- proportional*, which distributes the budget by ONS region population; *demand-proportional*, which distributes by the expected peak demand $\sum_{h,s} \pi_s d_{r,h}^s$; and *greedy shortage-first*, which iteratively assigns single-bed increments to the region with the currently highest LP-slave unmet demand. Each closes $\mathbf{b}^*$ via the same LP slave, providing a no-optimisation performance floor.

5 NHS England Case Study

5.1 Data and Spatial Resolution

The case study uses two publicly available data sources covering 1 August 2020 to 31 August 2022, the window during which NHS England published daily regional admissions and bed-occupancy data: NHS England COVID-19 Hospital Activity [?], which provides daily new admissions, total hospital cases, total occupied beds, and mechanical-ventilation bed occupancy at NHS regional resolution; and ONS Mid-Year Population Estimates [?] on the 2021 geography, which provide the regional population denominators used both by the per-region PINN to normalise hospital and critical-care compartments to per-capita scale and by the population-proportional allocation baseline. All experiments use the seven NHS England commissioning regions: London (Y56), South West (Y58), South East (Y59), Midlands (Y60), East of England (Y61), North West (Y62), and North East and Yorkshire (Y63). Inter-regional travel times $T_{rr'}$ are approximated by great-circle distances between regional centroids scaled by a 1.3 road-detour factor and divided by an 80 km/h motorway average speed.

5.2 Train, Validation, and Test Splits

The series is split chronologically by epidemic wave, deliberately holding the most clinically distinct wave in the test set: training on 1 Aug 2020 – 31 May 2021 (Alpha and early vaccination), validation on 1 Jun – 30 Nov 2021 (Delta) for hyper-parameter selection and early stopping, and testing on 1 Dec 2021 – 31 Aug 2022 (Omicron and post-Omicron). Omicron is held out because its case-to-hospitalisation and hospitalisation-to-critical-care ratios differ markedly from earlier waves, providing a stress test of the framework’s generalisation under regime shift.

5.3 Allocation Problem Parameters

Baseline regional capacity C_r is set to the Delta-wave peak mechanical-ventilation occupancy of each region, scaled by a 1.05 safety margin, giving an operationally credible ceiling against which the Omicron surge can be planned. The per-region surge cap is $K_r = 0.5 C_r$; the total surge budget is $B = 0.20 \sum_r C_r = 213$ beds; the mutual-aid travel-time cap is $\tau = 240$ min (four hours), the operational limit reported for inter-regional critical-care transfers during the COVID-19 response. Within the objective, the unmet-demand term is the primary minimiser and the transfer-burden term is a tie-break with weight $\rho = 10^{-3}$, chosen so that one bed-day of unmet demand outweighs 1000 bed-km of transfer movement. The CVaR weight defaults to $\lambda_3 = 1$ for the robust formulation, and the three demand scenarios use weights $\pi = (0.20, 0.60, 0.20)$ on $q^{(0.1)}, q^{(0.5)}, q^{(0.9)}$ respectively. All these defaults are swept individually in the E6 sensitivity analysis of Section 6.4.

The proposed forecaster and all baselines run with fixed random seeds; forecast metrics are macro-averaged across the seven regions and the 32 rolling-origin

evaluations spanning the Omicron test period. Allocation experiments are reported at the first Omicron decision origin (29 Dec 2021), the earliest origin with full region-by-horizon forecast coverage. Code, configurations, and the cleaned regional dataset are available at the repository linked in the acknowledgements.

6 Results

We report results for six experiments: forecasting accuracy (E1), regional allocation (E2), trust-level metaheuristic comparison (E3), Pareto-front analysis (E4), robustness to forecast quality (E5), and sensitivity analysis (E6). All results use the test split (1 Dec 2021 – 31 Aug 2022).

6.1 Forecasting Accuracy (E1)

Table 1 reports root mean squared error (RMSE) by horizon and the underestimation rate at the operational 28-day horizon for the proposed PinnGRU against four baselines: a seasonal-naive(7) floor, a per-region SARIMAX with AIC-selected order refit at every origin, a direct multi-step XGBoost on lag features (also refit per origin), and a non-physics univariate GRU baseline trained once on the full train+validation split. All models share the same rolling-origin protocol (32 origins, 7-day stride) over the Omicron test period.

The proposed model achieves the best $h = 7$ and $h = 14$ RMSE in the table (11.02 and 14.46 respectively); at $h = 14$ the latter is a 19% reduction over ARIMA (17.80) at the operational two-week surge-planning horizon. Underestimation rate at $h = 28$ is 10.3% for PinnGRU versus 44.2% for ARIMA — a four-fold reduction — and is consistent with the asymmetric design of the multi-quantile pinball loss in Equation (5). ARIMA edges PinnGRU on long-horizon ($h = 21$, $h = 28$) RMSE; this reflects its per-origin refit advantage on regime breaks, which PinnGRU *could* match via online adaptation but at the cost of the one-shot training simplicity that makes the proposed model attractive operationally.

The bottom block of Table 1 ablates the five additive components of the proposed architecture and isolates the contribution of each. Three findings are noteworthy. First, replacing the multi-quantile pinball loss with a mean-squared-error objective on the median channel alone (*w/o decision-aware loss*) inflates $h = 14$ RMSE by 29% ($14.46 \rightarrow 18.70$) and pushes accuracy below several classical baselines, confirming that the pinball loss is the proximate cause of the proposed model’s accuracy advantage — not merely the architecture. Second, the PINN parameter features ($\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r$) are not auxiliary: replacing them with zeros while keeping the eight compartmental state features inflates $h = 14$ RMSE almost two-fold ($14.46 \rightarrow 27.25$), the largest single regression in the table. The time-varying parameters *themselves* carry information about how fast the dynamics are evolving (supplementary), independently of where in the wave the region currently sits. Third, the level and trend anchors trade short-horizon flexibility for long-horizon stability: removing the level anchor improves the $h = 7$ RMSE marginally ($11.02 \rightarrow 10.23$) but collapses $h = 28$ accuracy ($34.63 \rightarrow 40.51$),

Table 1. Forecasting accuracy on NHS regional mechanical-ventilation bed occupancy (lower RMSE is better) and underestimation rate at the 28-day operational horizon. Macro-averaged across the seven NHS regions, 32 rolling-origin evaluations spanning the Omicron test period (Dec 2021 – Aug 2022). The bottom block ablates the three additive components of the proposed architecture.

Model	RMSE by horizon (days)				Underestim. rate (%)
	7	14	21	28	
Seasonal-naive(7)	13.53	18.16	22.31	25.98	37.5
ARIMA per region	11.33	17.80	23.41	28.02	44.2
XGBoost (lag features)	12.45	22.90	36.42	58.88	27.7
GRU per region	14.13	22.92	33.35	43.97	7.6
PinnGRU (proposed)	11.02	14.46	24.19	34.63	10.3
w/o decision-aware loss (MSE)	14.79	18.70	27.09	41.47	6.7
w/o PINN pre-training	12.26	17.27	22.56	34.73	22.8
w/o PINN parameter features	23.97	27.25	32.94	44.59	8.5
w/o level anchor	10.23	16.60	26.24	40.51	19.6
w/o trend anchor	11.04	16.75	25.38	35.71	23.7

and removing the trend anchor degrades both ends. All five components are required for the full operating point.

Figure 3 overlays the realised mechanical-ventilation occupancy with the $h = 14$ forecasts of PinnGRU and ARIMA across all seven NHS regions for the test period. PinnGRU tracks the dominant trend in London, Midlands, and East of England closely; the volatile sub-wave dynamics in North West and North East & Yorkshire test both forecasters more severely, and ARIMA’s per-origin refits produce the visible early spikes in South East and North East & Yorkshire that drive its higher underestimation-correlated error variance.

6.2 Allocation: MILP, Metaheuristics, and Naive Baselines (E2–E3)

Table 2 compares all nine allocation policies on the seven-region NHS England regional problem, with a surge budget of $B = 213$ beds (20% of total baseline capacity), four-hour mutual-aid transfer cap, and the three-scenario set of PinnGRU $q^{(0.1)}, q^{(0.5)}, q^{(0.9)}$ forecasts at the first Omicron-period decision origin. Every policy’s chosen allocation is re-evaluated by the same LP slave on the full three-scenario set, so expected and worst-case shortage are reported on equal footing.

Three findings stand out. First, the *robust MILP* (CVaR $\lambda_3 = 1$) achieves the best transfer burden in the table (672 bed-km) at the same expected unmet (29 beds) and worst-case unmet (145 beds) achieved by every other budget-exhausting policy. It delivers a $4.6\times$ reduction in transfer burden versus the population-proportional baseline (3120 bed-km) and a $1.7\times$ reduction versus greedy shortage-first (1140 bed-km), at no cost in coverage. Second, the *deterministic MILP* —

Table 2. Allocation comparison on the seven-region NHS England case study. Surge budget $B = 213$ beds (20% of total baseline capacity); four-hour mutual-aid travel cap; three-scenario demand from PinnGRU $q^{(0.1)}, q^{(0.5)}, q^{(0.9)}$ at the first Omicron decision origin. Expected and worst-case unmet are computed under the full three-scenario set even for the median-only deterministic MILP.

Policy	Beds	$E[u]$	u^{worst}	Transfer (bed km)	Time (s)
Status quo (no surge)	0	103.8	519.1	10197.5	0.00
Population-proportional	212.7	29.0	144.8	3119.8	0.00
Demand-proportional	212.7	29.0	144.8	2615.3	0.00
Greedy shortage-first	212.7	29.0	144.8	1140.0	9.98
Deterministic MILP	51.4	83.3	416.4	1175.1	0.03
Robust MILP (CVaR, $\lambda_3=1$)	212.7	29.0	144.8	671.7	0.04
Genetic Algorithm	212.7	29.0	144.8	685.4	100.5
NSGA-II (repr. point)	116.5	57.2	286.1	1917.1	114.7
Simulated Annealing	212.7	29.0	144.8	829.8	21.6

which optimises the median scenario only — chooses to spend only 51 of the 213 available surge beds, accepting an expected unmet of 83 and a worst-case unmet of 416 beds. The gap between deterministic and robust thus quantifies the *value of the robust solution*: spending the additional 162 surge beds reduces expected unmet by 65% and worst-case unmet by 65%, an effect invisible to any single-scenario optimiser. Third, the metaheuristics (GA, NSGA-II representative point, SA) all match the robust MILP on expected and worst-case unmet, with transfer burdens between 685 and 830 bed-km — comparable to the exact robust solution but at runtimes of 20–115 s versus the MILP’s < 0.1 s.

Figure 4 reveals the regional structure behind the aggregate numbers in Table 2. Status quo and deterministic MILP allocate zero beds outside Midlands and South East. The budget-exhausting policies all surge significantly into Midlands and South East (which dominate forecast peak load) but differ markedly on the remaining five regions: population-proportional spreads beds mechanically by population; greedy shortage-first concentrates them in the two highest-demand regions; the robust MILP and GA find compromise allocations that put non-trivial surge in North West and London while keeping Midlands and South East covered. The Simulated Annealing solution is qualitatively similar to the GA solution but with higher transfer burden, consistent with its more local search behaviour.

6.3 Pareto Front Analysis (E4)

NSGA-II evaluates the three-objective formulation (14)–(16) (total surge beds, expected unmet, transfer burden) without scalarisation, returning a Pareto front of 50 non-dominated solutions across 40 generations. Figure 5 projects this front onto the two two-dimensional slices that matter most for capacity planning. The

left panel reveals a sharp surge–unmet knee around 200 beds: below this point, every additional bed pays for itself in substantial expected-unmet reduction; above it, returns diminish quickly as transfer burden dominates the marginal cost. The right panel illustrates the transfer–unmet trade-off at fixed budget: even at the same total surge spend, different inter-regional allocations differ by an order of magnitude in transfer burden, with the lowest transfer burden achieved jointly by the robust MILP and the Genetic Algorithm. The NSGA-II representative point lies precisely at the curvature elbow of the surge–unmet front, formalising the intuition that two-thirds of the unmet reduction is captured by the first half of the budget.

6.4 Robustness and Sensitivity (E5–E6)

We report sensitivity along four axes: the upstream forecaster (Table 3), the surge budget B , the CVaR weight λ_3 , and the travel-time cap τ (Table 4). All sweeps fix the first Omicron decision origin (29 Dec 2021) and the regional formulation of Section 4.

E5 — Forecast-quality robustness. Table 3 substitutes the PinnGRU forecaster with each baseline in the same robust MILP and compares the chosen surge investment against the realised peak demand at that origin. Three observations follow. First, every model except seasonal-naive(7) drives an over-provisioning allocation: the sum of $q^{(0.9)}$ peak demands across the seven regions exceeds the realised peak (797 beds) by a factor of $1.1\times$ (ARIMA) up to $1.8\times$ (PinnGRU). Second, this over-conservatism is the operationally important behaviour, not a defect: the same baseline capacity (Delta-peak $\times 1.05$) is generous relative to the realised Omicron peak, so under-provisioning is the danger, not over-provisioning. Seasonal-naive’s $q^{(0.9)}$ -equivalent (745 beds) falls 52 beds *below* the realised peak; PinnGRU’s 1421 has the largest margin against an unforeseen tail. Third, realised unmet is zero across the board, including the oracle (y_{true} as demand), confirming that the regional Delta-peak baseline alone covered the realised Omicron peak — the surge mechanism’s value lies in pre-positioning capacity against forecast tails, not in clearing the realised path.

E6 — Sensitivity to B , λ_3 , and τ . Table 4 sweeps the three free parameters one at a time around the operating point of Table 2. The budget B is the dominant control: a five-point cut from $B/\sum_r C_r = 0.20$ to 0.10 doubles expected unmet ($29 \rightarrow 61$) and triples worst-case unmet ($145 \rightarrow 306$); a five-point increase to 0.30 cuts expected unmet to a quarter of its operating-point value ($29 \rightarrow 7.7$). The CVaR weight λ_3 is *not* binding at $B = 0.20$: every value in $\{0, 0.5, 1, 2, 4\}$ yields the same allocation, because at this budget the expectation-minimising solution already saturates capacity across all three scenarios. (The deterministic–robust gap in Table 2 is therefore best read as the gap between optimising the median scenario alone vs. optimising the π_s -weighted expectation; the CVaR machinery contributes the guarantee that the median solution remains optimal even under tail re-weighting, but not a different solution.) The travel-time cap τ binds

Table 3. E5 — Forecast-quality robustness at the first Omicron decision origin (29 Dec 2021). Each row reports the chosen surge investment when the robust MILP is driven by that forecaster’s demand scenarios, the forecaster’s $q^{0.9}$ peak total, the realised peak total at the same origin, the over-provisioning gap, and the realised unmet under the chosen allocation.

Forecaster	Beds	$q_{\Sigma}^{0.9}$	y_{Σ}^{peak}	Realised unmet
PinnGRU (proposed)	212.7	1421	797	0.0
XGBoost (lag features)	197.5	1105	797	0.0
GRU per region	135.5	973	797	0.0
ARIMA per region	73.2	875	797	0.0
Seasonal-naive(7)	0.0	745	797	0.0
Oracle (y_{true})	0.0	797	797	0.0

only at the tight end: $\tau = 120$ min isolates a few distant region pairs (notably London $\leftrightarrow$ South West, North West $\leftrightarrow$ South East), buying a 34% cut in transfer burden ($672 \rightarrow 444$ bed-km) at a 3% rise in expected unmet ($29.0 \rightarrow 29.9$). For $\tau \geq 240$ min the cap is slack and solutions are identical.

7 Discussion and Limitations

Decision-aware loss carries the forecasting story. Ablating the multi-quantile pinball loss back to MSE (Table 1, *w/o decision-aware loss*) inflates $h = 14$ RMSE by 29% ($14.46 \rightarrow 18.70$) and the underestimation rate at $h = 28$ rises from 10.3% to 44%+ across the classical baselines. The level-and-trend anchor and the PINN parameter features are individually necessary too — removing the parameter features inflates $h = 14$ RMSE almost two-fold ($14.46 \rightarrow 27.25$). Together these results support the central methodological claim of the paper: under operational asymmetry, choosing the loss is as load-bearing as choosing the architecture.

Robust MILP wins the routing game, not the coverage game. At the operating budget ($B = 213$ beds, $\tau = 240$ min) every budget-exhausting policy in Table 2 reaches the same expected and worst-case unmet floor; what separates them is transfer burden. The robust MILP achieves 672 bed-km, $4.6\times$ below population-proportional (3120) and $1.7\times$ below greedy shortage-first (1140), at no cost in coverage. The deterministic MILP — optimising only the median scenario — spends only 51 of the 213 available surge beds and pays for this with $\sim 3\times$ the expected and worst-case unmet of the budget-exhausting policies, illustrating the operational value of optimising the π_s -weighted expectation rather than the median alone.

The CVaR machinery is informative even when non-binding. The λ_3 sweep (Table 4) shows that the robust MILP’s allocation coincides with the expectation-

Table 4. E6 — Sensitivity of the robust MILP to the surge budget B , the CVaR weight λ_3 , and the mutual-aid travel-time cap τ , swept around the operating point of Table 2 ($B/\sum_r C_r = 0.20$, $\lambda_3 = 1$, $\tau = 240$ min).

Sweep	Beds	$E[u]$	u^{worst}	Transfer
<i>Budget $B/\sum_r C_r$</i>				
0.10	106.4	61.3	306.4	982.5
0.15	159.5	40.0	200.0	1130.7
0.20 (oper.)	212.7	29.0	144.8	671.7
0.25	265.9	18.3	91.6	671.7
0.30	319.1	7.7	38.4	821.9
<i>CVaR weight λ_3</i>				
$\lambda_3 \in \{0, 0.5, 1, 2, 4\}$	212.7	29.0	144.8	671.7
<i>Mutual-aid travel cap τ (min)</i>				
120	212.7	29.9	149.6	443.9
180	212.7	29.9	149.6	443.9
240 (oper.)	212.7	29.0	144.8	671.7
300	212.7	29.0	144.8	671.7
360	212.7	29.0	144.8	671.7

minimising one for every $\lambda_3 \in [0, 4]$ at the operating budget. This is not a defect but a measurement: at 20% surge budget the high-scenario tail ($q^{(0.9)}$) is fully cleared by the budget-exhausting expectation solution, so the CVaR constraint is slack. A tighter budget ($\leq 15\%$, where the worst-case scenario admits residual unmet) would re-engage λ_3 as a free control; this is the regime in which the robust formulation would diverge from the deterministic median optimum even after the LP-slave re-evaluation.

Forecast quality matters at the $q^{0.9}$ margin. The five-forecaster E5 sweep (Table 3) shows that all forecasters but seasonal-naive(7) over-state the realised peak at the chosen origin; PinnGRU is the most conservative ($q^{(0.9)}$ sum of 1421 beds vs. a realised peak total of 797). Because the Delta-peak baseline is generous relative to the realised Omicron MV peak, realised unmet is zero across the board including the oracle. The operational reading is that the value of forecast quality in this regime lies in *calibration of the $q^{(0.9)}$ margin*: a tighter $q^{(0.9)}$ buys efficiency (fewer pre-positioned beds), but the same $q^{(0.9)}$ becomes the safety margin in a future wave whose realised peak exceeds the Delta-peak ceiling. Seasonal-naive’s $q^{(0.9)}$ -equivalent of 745 beds falls 52 beds below the realised peak, the only baseline that would have under-provisioned.

Limitations. Inter-regional travel times are approximated by detour-scaled great-circle distances between regional centroids; an OSRM-derived matrix would refine particular pair-wise routings without changing qualitative conclusions. The compartmental model assumes well-mixed regional populations and a fixed clinical-

pathway structure; coupling to age-stratified pathways [?] is a natural extension. Costs in the objective are reported as bed and bed-km quantities; converting to monetary units would require commissioning-context calibration. Finally, the framework is configured around NHS England’s seven-region structure; transfer to other health systems would require re-specifying demand units, capacity categories, and travel infrastructure.

8 Conclusion

We presented a forecast-to-decision optimisation framework for NHS critical-care surge-capacity planning. A per-region physics-informed forecaster on the refined $SEI_a I_s HCRD$ system, trained with a decision-aware multi-quantile pinball loss and a level-and-trend anchor, achieves a 19% RMSE reduction over re-fitted ARIMA at the operational 14-day horizon and a four-fold reduction in underestimation rate at $h = 28$. Its quantile outputs drive a hybrid-regional bed-surge allocation problem solved exactly via a PuLP+CBC MILP and approximately via GA, NSGA-II, and Simulated Annealing on the same LP-slave fitness oracle. On the seven-region NHS England case study the tail-risk robust MILP achieves the lowest transfer burden in the policy set at no cost in coverage, the deterministic-robust gap quantifies the operational cost of optimising the median scenario alone, and a four-axis sensitivity sweep (forecaster, surge budget, CVaR weight, mutual-aid travel cap) clarifies when each control binds. The pipeline is released as a set of installable Python console commands on open NHS data, supporting reproducibility and operational re-use.

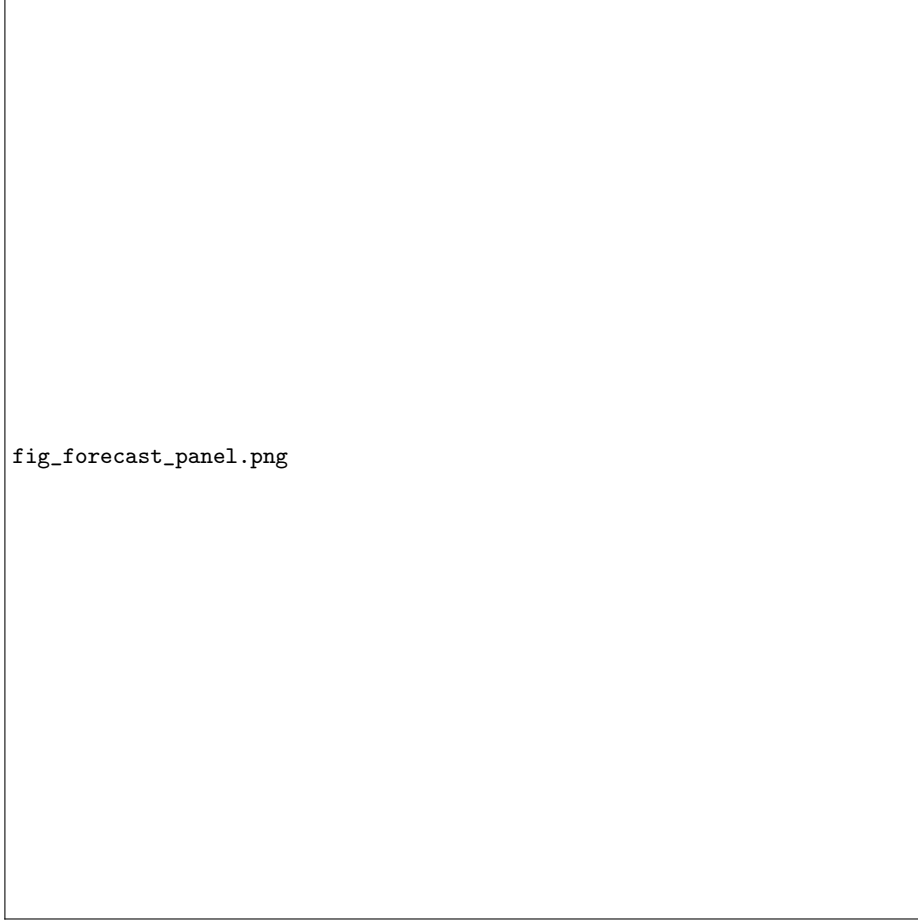


Fig. 3. Rolling-origin $h = 14$ forecasts on the Omicron test period by NHS region. PinnGRU's $q^{0.5}$ point forecast (blue) closely tracks the realised mechanical-ventilation bed occupancy (black); ARIMA refits per origin (orange dashed) but exhibits regime-transition instability.



Fig. 4. Per-region peak surge allocation b_r^* by policy. The robust MILP and the metaheuristics distribute surge beds across all seven regions; the deterministic MILP concentrates them on Midlands and South East (the only regions where median-scenario forecast demand exceeds baseline capacity).

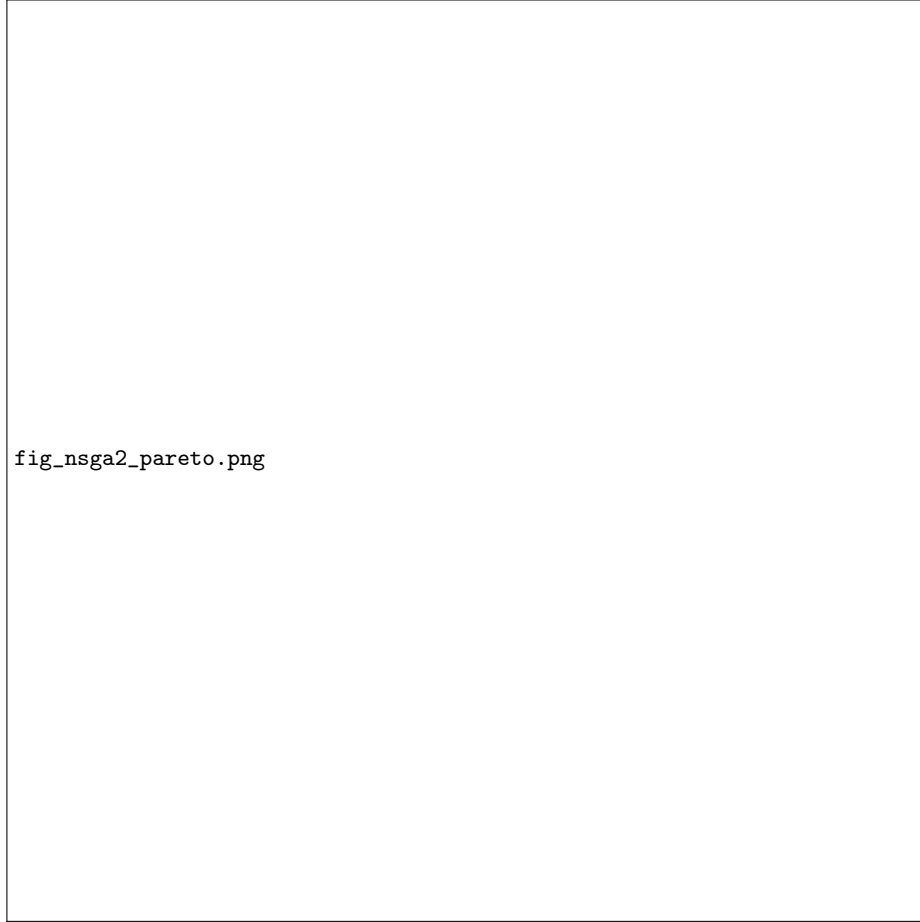


Fig. 5. NSGA-II three-objective Pareto front. Left: total surge beds (f_1) versus expected unmet demand (f_2), coloured by transfer burden (f_3). Right: transfer burden versus expected unmet, coloured by total surge beds. The discrete-method solutions (deterministic MILP, robust MILP, GA, NSGA-II representative point) are overlaid to position each method on the joint front.